Simple AAC Sri Lanka — Complete User Guide (v1.1.0)

Your Comprehensive Reference Manual for Daily Communication, Customization, and Accessibility

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Chapter 1 — Getting Started with Simple AAC

Simple AAC Sri Lanka is an augmentative and alternative communication (AAC) platform engineered specifically for non-verbal children, individuals with autism (ASD), speech delays, aphasia, or stroke recovery. Built with an offline-first philosophy, Simple AAC provides instantaneous speech playback in **English, Sinhala, and Tamil** without requiring any internet connection or cloud subscription.

1.1 Main Interface Overview

When you launch Simple AAC, you are greeted by the clean, calming default Blue Theme designed to maximize readability and reduce sensory overstimulation. The interface consists of three primary areas:

- **Top Application Bar:** Houses the **Pin Board** security lock (left), **Simple AAC** title (center), and the **Menu (Drawer)** button (right).
- **Sentence Composer Bar:** Displays selected cards in sequence with immediate speech controls (Delete, Clear, Speak).
- **AAC Card Grid:** High-contrast visual communication cards designed for easy targeting by fine-motor challenged fingers.

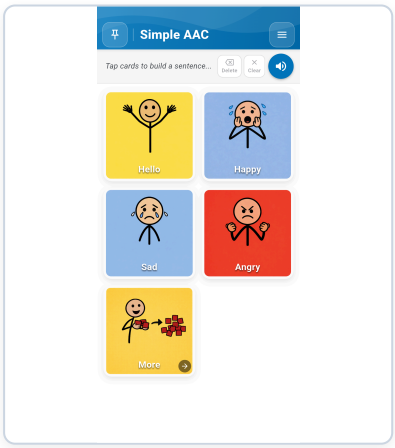


Figure 1.1: Home Screen with Blue Header & Sentence Bar

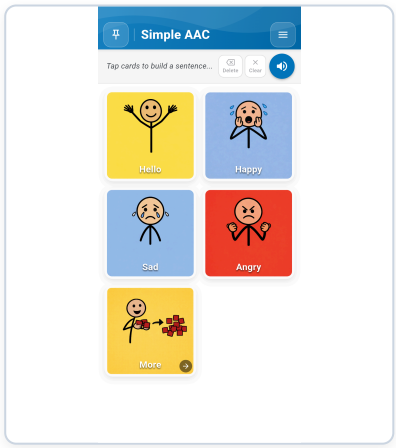


Figure 1.2: High-contrast AAC Communication Cards Grid

Chapter 2 — Speaking and Sentence Composition

2.1 Tapping to Speak

Communication begins with a single touch. Simple AAC provides immediate multi-sensory feedback:

- **Instant Speech:** When *Tap to Speak* is enabled in Settings, touching any card speaks its label immediately via offline Text-to-Speech (TTS) or custom recorded voice.
- **Haptic Pop:** Every touch triggers a responsive physical vibration and subtle visual pop effect, confirming selection for tactile learners.

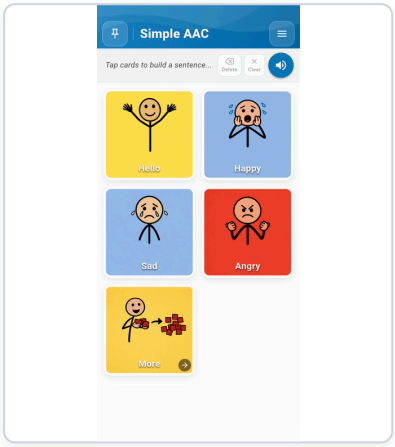


Figure 2.1: Live demonstration of tapping cards & speech

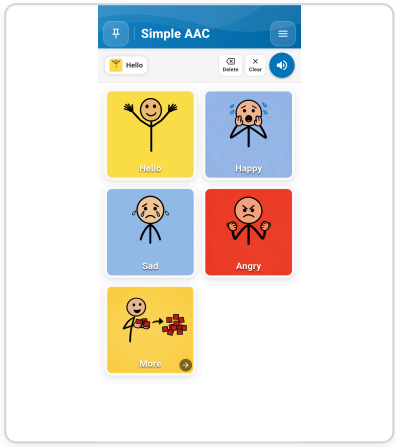


Figure 2.2: Tapping "Hello" places card on sentence bar

2.2 Constructing Multi-Word Sentences

Users can express complex thoughts by tapping cards sequentially (e.g., *"Hello"* + *"Happy"*). The Sentence Composer Bar provides three primary actions:

- **Delete (Back Arrow):** Removes the most recently added card from the sentence.
- **Clear:** Empties the entire sentence bar in a single tap.
- **Speak Sentence (Speaker Button):** Speaks the entire constructed sentence fluently from left to right.

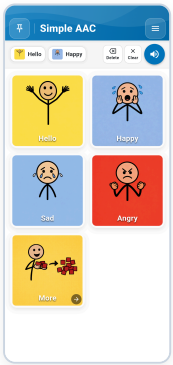


Figure 2.3: Building multi-word sentences in the Sentence Composer Bar

Chapter 3 — Customizing Cards and Multilingual Content

Every individual communicates differently. Simple AAC allows caregivers, speech therapists, and educators to fully personalize every card's imagery, labels, and audio.

3.1 Enabling Edit Mode & Card Editor

To customize cards, tap **Menu** (top right) → **Enable Edit Mode**. Red trash-can badges appear on cards, and an **Add New** button displays at the top. Tapping any card opens the comprehensive Edit Card Dialog.

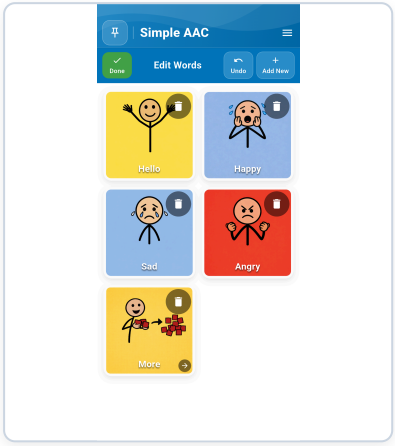


Figure 3.1: Edit Mode active with Add New button

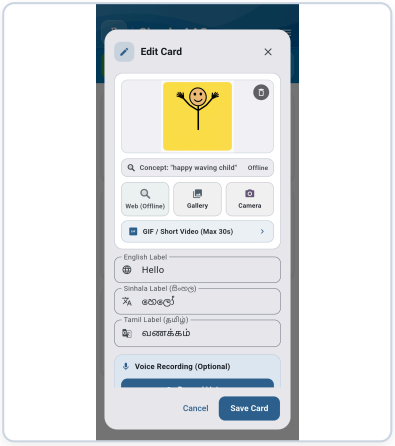


Figure 3.2: Card Editor Dialog with customization tools

3.2 Adding Authentic Photos & Smart Search

- **Camera:** Capture real-world items (child's bottle, favorite toy, family members). The app requests standard Android permissions automatically.
- **Gallery:** Select family photos or therapy pictograms stored on the device.
- **Smart Image Search:** When online features are enabled, search a curated, child-safe library of concepts.

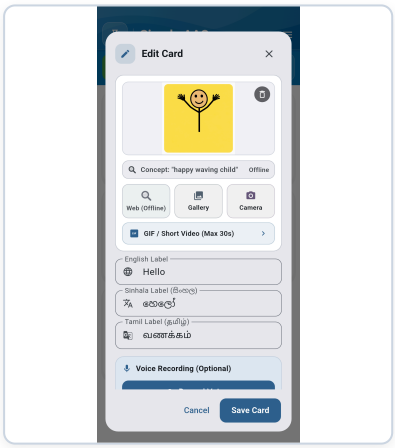


Figure 3.3: Direct Camera photo capture

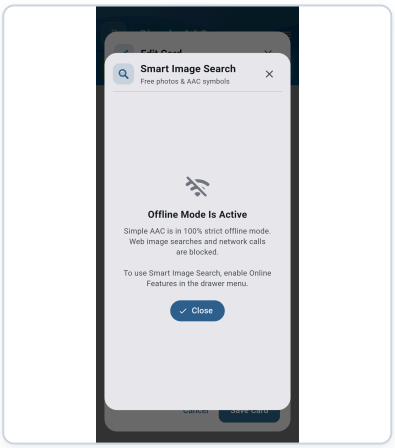


Figure 3.4: Child-safe Smart Image Search

3.3 Trilingual Support & Custom Voice Recording

Simple AAC supports Sri Lanka's official languages with simultaneous trilingual labels: **English** (e.g., *Hello*), **Sinhala** (*හෙලෝ*, *සතුටයි*, *වතුර*), and **Tamil** (*வணக்கம்*, *மகிழ்ச்சி*, *தண்ணீர்*). For children who respond best to familiar family voices, caregivers can record custom audio directly in the editor, which takes precedence over synthesized speech.



Figure 3.5: Trilingual English, Sinhala & Tamil labels



Figure 3.6: High-fidelity custom voice recording

Chapter 4 — Animated GIFs and Video Loops

Visual learners and individuals with autism frequently comprehend dynamic actions (such as *jumping*, *drinking*, or *washing hands*) significantly better through motion than static images. Simple AAC includes a dedicated high-performance GIF engine that loops smoothly without draining battery power.

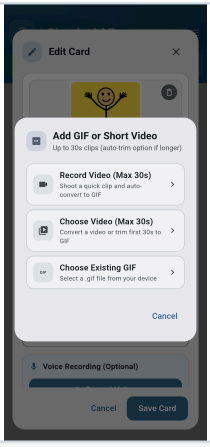


Figure 4.1: GIF import for loopable action cards

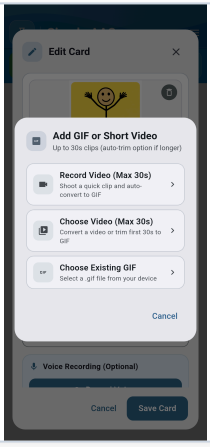


Figure 4.2: Built-in 30s video-to-GIF converter

Caregivers can record short video clips (up to 30 seconds) of actual therapies, routines, or sign language gestures, and the app automatically converts them into lightweight, battery-efficient animated loops.

Chapter 5 — Folders and Category Organization

As vocabulary expands, cards can be organized into categories such as *Feelings*, *Food & Drink*, *School*, and *Activities*. Tapping a folder card smoothly navigates into the category deck while keeping the sentence composer bar persistent at the top.

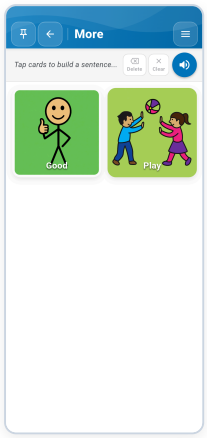


Figure 5.1: Navigating into subfolders and category decks ("More ->")

Chapter 6 — Appearance, Themes and Haptic Feedback

6.1 Settings Drawer & Live Color Customization

Access the Settings Drawer by tapping the top-right Menu button. Simple AAC v1.1.0 introduces redesigned visual Color Pickers equipped with real-time interactive preview canvases.

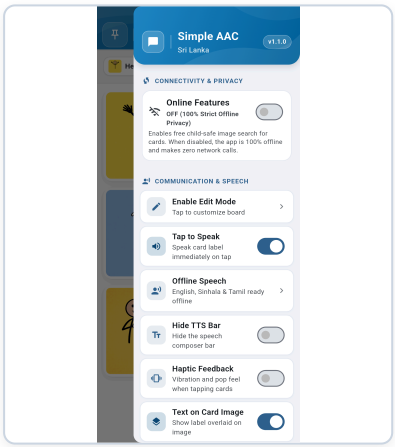


Figure 6.1: Settings Drawer with haptics toggle

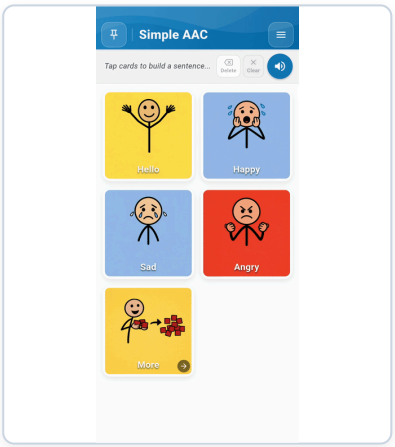


Figure 6.2: Live Color Picker demonstration

6.2 Redesigned Color Pickers

Customize the primary app color, global text color, and board background with real-time contrast previews:

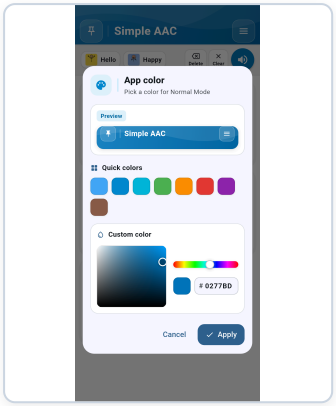


Figure 6.3: App Color Picker with mini-header

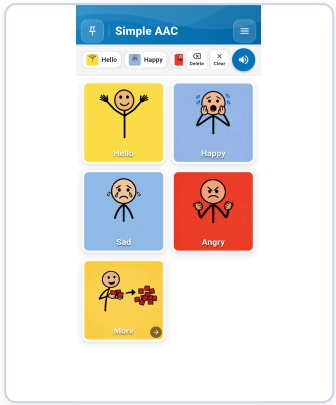


Figure 6.4: Text Color Picker with 3-card preview

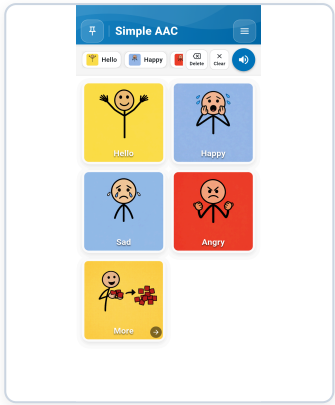


Figure 6.5: Board Background Color preview

6.3 Color Vision Support & Haptic Feedback

For users with color vision deficiency, Simple AAC provides a tested **Color-Blind Safe Mode** exceeding WCAG AAA contrast standards. The **Haptic Feedback** switch delivers tactile reinforcement on every tap.

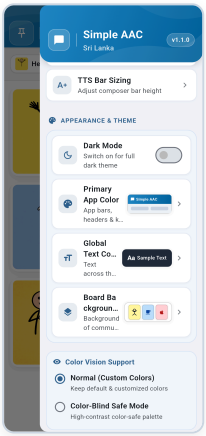


Figure 6.6: Color Vision Support and high-contrast settings

Chapter 7 — Safety, Board Pinning and Child Lock

7.1 Enabling Board Pinning with a PIN

Accidental taps or exploration can inadvertently alter settings or delete cards. Tap the **Pin Board** button (top left) to set a 4-digit security PIN. When locked, the Settings Drawer and Edit Mode are completely disabled, allowing users to communicate safely without risking data alteration.

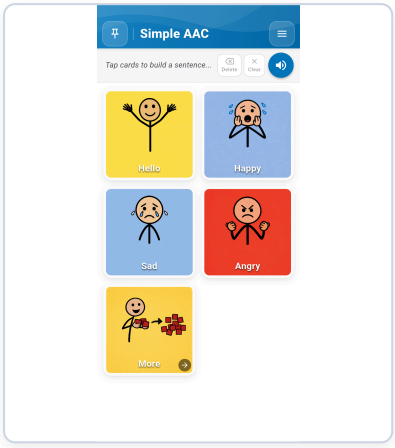


Figure 7.1: Setting up or unlocking 4-digit security PIN

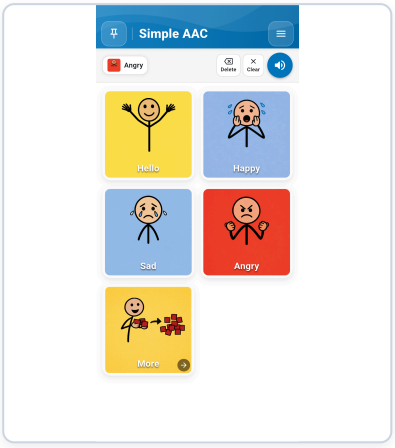


Figure 7.2: Pinned board mode with disabled edit controls

Chapter 8 — Data Backup, Restore and Offline Privacy

8.1 Strict Offline Privacy Model

Simple AAC operates entirely on-device. Personal photos, voice recordings, and communication logs are never transmitted to external servers.

8.2 Exporting & Restoring Backups

- **Export Backup:** Packages all custom cards, photos, recordings, and settings into a single ZIP archive saved directly to your device's `Downloads` folder.
- **Import Restore:** Safely unpacks and validates any previously exported Simple AAC ZIP archive, instantly restoring your personalized vocabulary.

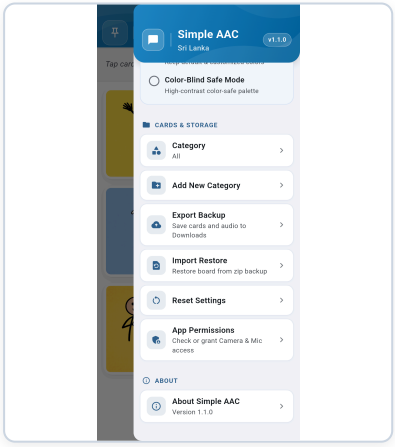


Figure 8.1: Cards & Storage section in Settings Drawer

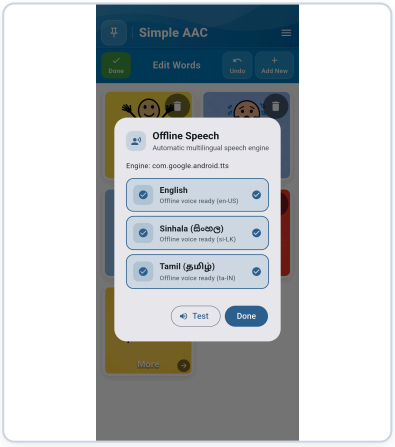


Figure 8.2: Offline Speech (TTS) engine configuration

> **Note on Scalability:** Simple AAC utilizes an atomic file persistence architecture (`aac_cards_v1.json` with automated `.bak` redundancy). Backups are verified safe for large datasets up to 1,000+ cards with full media integrity.

Simple AAC Sri Lanka • Version 1.1.0 • Empowering voices with dignity, independence, and reliability.